A bounded Alpoge–Sussman peek: what returns to Newton in 0+1

Result and source comparison, 2026-10-27. The useful immediate imports are a connected-response test for interaction, retention of the complete density through blocking, and reflection positivity as a condition that survives limits. None of these alone selects a positive $\hbar$. In our quartic insertion law, an exact moment identity gives a strictly negative fourth connected response at every positive $\hbar$. Its raw magnitude vanishes as a time cell shrinks, while the correctly scaled quartic response retains the original interaction.

The user authorized this bounded peek after the Newton work and status checkpoint were pushed. The source is ethanwsussman/Yang–Mills-LLM-experiment, pinned at `52cee131db140d7467cab4a263a1f04aa15a260f`. The source companion records the downloaded files and exact reading coverage. These are selected passages, not a proof review of QED3, YM3 or YM4. The repository itself labels the manuscripts tentative and not satisfactorily vetted.

Retrieval for the calculation: the nonlinear time cell supplies the normalized full insertion law and its moment bounds; the time-blocking note separates finite cell control from a continuum scale; and refinement composition keeps surviving physical information distinct from geometric convergence. The finite identities below are written derivations, unrefereed, with no literature priority claim.

Targeted check (GPT-6.1 Terra, 2026-10-27). Terra checked sections 2–4, including the strict cumulant sign, all $\hbar$ powers, the T^5 coefficient and the Gaussian reflection inequality at zero. Its test-function domain clarification is incorporated below. It did not review the external manuscripts or physical selection premises.

1. What is actually in the current source

The current README describes separate QED3 and YM3 manuscripts and a rotation argument called RP1. It continues to describe YM4 as work being developed and digested. This is an existence programme; this peek establishes no mass-gap claim. The following statements are the source's claims, at the indicated reading levels.

- **RP1, statement and selected argument passages read.** Translation invariance, an exponential moment bound, and reflection positivity across coordinate and diagonal mirrors are claimed to imply $O(d)$ covariance of separated-point correlations for $d \geq 2$ and tensor rotation weights of absolute value at most two. The conclusion allows contact terms at coincident points. Its OS Hilbert-space passage constructs a nonnegative time generator; the later rotation argument uses a second direction, analytic continuation and angular Fourier coefficients. It is not a theorem that positivity in time alone produces spacetime rotation symmetry. The root README describes RPM1 as a hybrid exposition partly written by Ethan Sussman and completed by Astra; it does not specify an Astra version or attribute all of the Yang–Mills work to that model.
- **PR1, abstract and two theorem statements read.** A compact probe has a positive fourth logarithmic determinant response. The proposed lattice comparison separates a cutoff error, uniform in torus size, from an exponentially small volume error. This is a concrete nontriviality witness. Its sign belongs to its fermion determinant observable; it is not the sign asserted for our scalar moment below.
- **NL1, abstract, setup and selected theorem clauses read.** The nonlinear step keeps local activities, the mass and vacuum terms, and an independent external-source scalar term. It claims an exact outgoing density and an unnormalized restricted marginal, under explicit primitive and domain hypotheses. The methodological lesson is to keep the source and scalar terms when differentiating or iterating the block map.

The dated October 23 revision list was also read. It reports local corrections, including support conventions, normalization and the placement of folded cyclic traces. Its assertion that conclusions are unchanged is not an independent verification here.

2. A connected fourth response in the exact Newton cell

Use the convex quartic cell at zero source, with

$$d\nu(z) = Z^{-1} \exp\left[-\frac{1}{\hbar}\left(\frac{a}{2}z^2 + \frac{b}{4}z^4\right)\right] dz, \qquad a, b, \hbar > 0, \qquad m_j = \mathbb{E}_\nu Z^j. \tag{1}$$

Here $a = mT/(uv)$ and $b = w\lambda$, with $T = u + v$ and $w = T/2$. The coefficient b has units action/length^4. Symmetry gives zero odd moments, and the quartic tail justifies all differentiations and integrations by parts below.

Integration by parts against z and z^3 gives

$$am_2 + bm_4 = \hbar, \qquad am_4 + bm_6 = 3\hbar m_2. \tag{2}$$

Multiply the second identity by m_2, the first by m_4, and subtract:

$$\boxed{m_4 - 3m_2^2 = -\frac{b}{\hbar}\left(m_2 m_6 - m_4^2\right) < 0.} \tag{3}$$

Cauchy–Schwarz gives $m_4^2 \le m_2 m_6$ by comparing Z and Z^3. Equality would require Z^3 proportional to Z almost surely, which is impossible for this everywhere positive density. Thus (3) is a strict finite test of its departure from a Gaussian law. No small-b or small-$\hbar$ expansion was used.

For a source j of units action/length, let $N(j) = \mathbb{E}e^{jZ/\hbar}$. Then

$$(\log N)^{(4)}(0) = \frac{m_4 - 3m_2^2}{\hbar^4}, \qquad N^{(4)}(0) = \frac{m_4}{\hbar^4}. \tag{4}$$

The subtraction matters: an ordinary fourth moment is nonzero even for a free Gaussian. A connected response removes the three Gaussian pairings. For the retained action of the insertion cell,

$$A_\hbar(\mu) - A_\hbar(0) = \frac{a}{2}\mu^2 - \hbar \log N(a\mu), \qquad A_\hbar^{(4)}(0) = -\frac{a^4}{\hbar^3}(m_4 - 3m_2^2) > 0. \tag{5}$$

This uses the same normalization and source convention as the nonlinear-cell note. It is an additional full-law response identity, not an inference from the background trajectory alone.

3. Refinement can hide an interaction in an unscaled probe

Put $g = b\hbar/a^2$ and $Y = \sqrt{a/\hbar}\, Z$. The normalized law of Y is proportional to $e^{-Y^2/2 - gY^4/4}$. As $g \downarrow 0$, dominated convergence gives its second, fourth and sixth moments $1, 3, 15$. Equation (3) therefore gives

$$m_4 - 3m_2^2 \sim -\frac{6b\hbar^3}{a^4}, \qquad A_\hbar^{(4)}(0) \sim 6b. \tag{6}$$

For equal time pieces $u = v = T/2$, $a = 4m/T$, $b = \lambda T/2$ and $g = \lambda\hbar T^3/(32m^2)$. At fixed $m, \lambda, \hbar > 0$, shrinking the cell consequently gives

$$m_4 - 3m_2^2 \sim -\frac{3\lambda\hbar^3}{256m^4}T^5, \qquad \frac{A_\hbar^{(4)}(0)}{w} \longrightarrow 6\lambda. \tag{7}$$

Thus the raw local cumulant tends to zero even though the retained potential's fourth derivative tends to the nonzero derivative of $\lambda\mu^4/4$. Neither a finite nonzero cell cumulant nor its raw

vanishing establishes the status of an interacting continuum theory. A fixed macroscopic observable and an estimate uniform in refinement are required. That is the useful parallel with PR1's cutoff and volume comparison, without importing its unreviewed field-theory proof.

4. The 0+1 reflection-positivity control, including zero action

RP1's rotation theorem needs $d \geq 2$. Its time-positivity ingredient has a direct $0+1$ control. Fix $m, \omega > 0$ and consider a centered stationary Gaussian process with covariance

$$C_\hbar(t-s) = \frac{\hbar}{2m\omega} e^{-\omega|t-s|}, \qquad \hbar \geq 0. \tag{8}$$

At zero multiplier this is the deterministic zero process. For real smooth compactly supported positive-time tests f_i, write $Q(f) = \iint f(t) C_\hbar(t-s) f(s) dt\, ds$ and $u_i = \int_0^\infty f_i(t) e^{-\omega t} dt$. Time reflection is $(\theta f)(t) = f(-t)$. Its characteristic functional obeys

$$\Phi(f_i - \theta f_j) = e^{-Q(f_i)/2} e^{-Q(f_j)/2} \exp\left(\frac{\hbar}{2m\omega} u_i u_j\right). \tag{9}$$

For any finite collection, the last kernel is positive semidefinite: expand its exponential into the sum of rank-one kernels $[\hbar/(2m\omega)]^n u_i^n u_j^n / n!$. Multiplication by the displayed positive diagonal factors preserves positivity. This proves the full Gaussian exponential-observable reflection inequality, including at $\hbar = 0$, where every matrix entry is one.

Hence reflection positivity and a time generator bounded below do not exclude a trivial zero-action member. Positive $\hbar$ in (8) is a supplied covariance scale; the quartic witness (3) is nonzero for each positive value but collapses at zero. No universal positive action value or relativistic c follows from these tests.

5. Consequence for STATE

The authorized Alpoge–Sussman peek is recorded at a fixed commit with selected source coverage. It contributes a concrete fourth-response test and a warning to preserve complete source-dependent densities. The main Newton queue remains physical justification of preparation and record closure and identification of its positive scale with $\hbar$. The central-limit preparation test listed in STATE remains unstarted. No YM proof claim, gap construction or rotation theorem has been imported as an established result.